

12 (long)

(a) A thin film characterised by Young's modulus E , Poisson's ratio ν , thermal expansion coefficient α_1 and thickness t is perfectly bonded to a component with a significantly larger thickness and a different thermal expansion coefficient α_2 , as shown in Fig. 8(a). The material system undergoes a temperature increase of ΔT .

(i) Derive an expression for the strains ε_x and ε_y in the film following this temperature change. [6]

(ii) Using your answer from part (i) and assuming $\sigma_z = 0$, derive an expression for the stresses σ_x and σ_y in the film. [9]

(b) An aluminium bar and a steel bar are firmly attached to the opposing sides of a rigid plate, which is thin compared to the length of the bars. The assembly is fixed between two rigid supports, as shown in Fig. 8(b). The aluminium bar is 15 m long with a cross-sectional area of 2 m^2 , and the steel bar is 10 m long with a cross-sectional area of 3 m^2 . Aluminium has a Young's modulus of 68 GPa and a thermal expansion coefficient of $22 \times 10^{-6} \text{ K}^{-1}$, while steel has a Young's modulus of 210 GPa and a thermal expansion coefficient of $12.5 \times 10^{-6} \text{ K}^{-1}$. Assume the bars behave elastically without buckling and ignore the self-weight of the system. Initially at 20°C , the assembly is stress free.

(i) An external force of $F = 5000 \text{ kN}$ is applied to the rigid plate at 20°C . Find the stresses in the Aluminium bar and the Steel bar. [6]

(ii) The assembly is subsequently heated from 20°C to 80°C , with the external force F still applied. Find the stresses in the aluminium and the steel bars after the temperature change. [9]

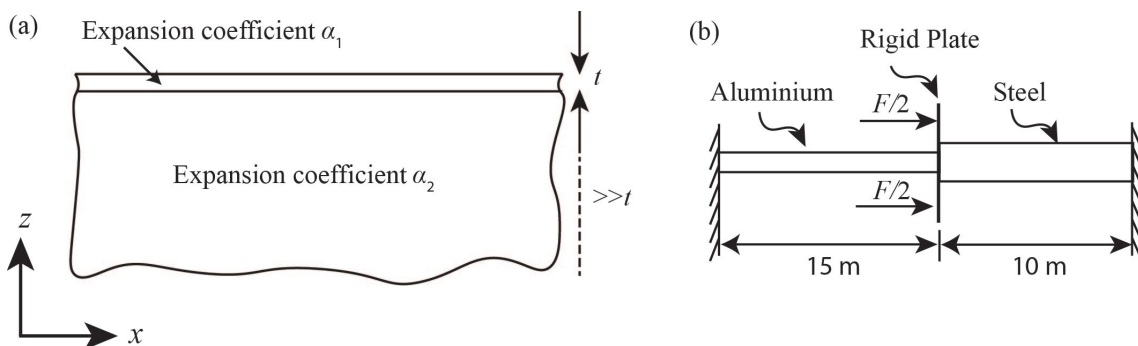


Fig. 8

END OF PAPER